



Sec: Sr. IIT

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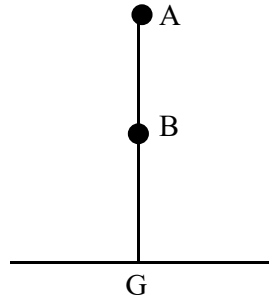
Daily Practices Sheet - 09

PHYSICS

1. A particle is dropped from rest from point A and B is another point on the downward path of the particle. Time taken by the particle during its journey from A to B is

$\left[\frac{1}{(\sqrt{7}-1)} \right]^{\text{th}}$ of the time taken by particle during its journey from B to the ground G.

If $AG = h$, find path length of AB.



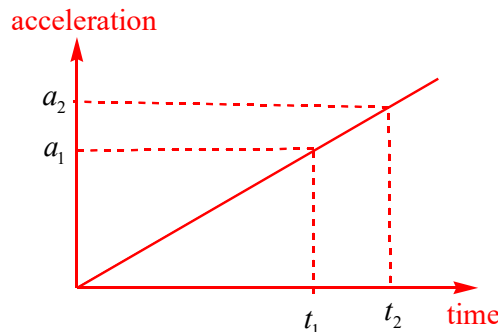
1) $\frac{h}{7}$

2) $\frac{h}{\sqrt{7}}$

3) $\frac{h}{(\sqrt{7}-1)}$

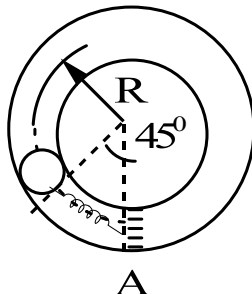
4) $\frac{h(\sqrt{7}-1)}{(\sqrt{7}+1)}$

2. Acceleration-time graph of a particle is shown. Work done by all the forces acting on the particle of mass m in time interval t_1 and t_2 , while a_1 is the acceleration at time t_1 , is given by:

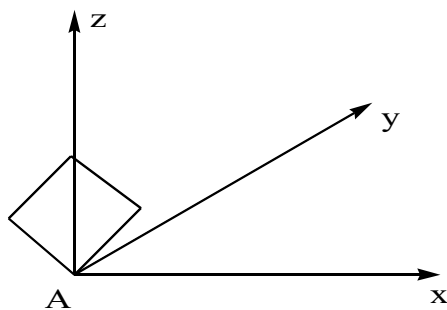


1) $\frac{ma_1^2}{4t_1}(t_2^3 - t_1^3)$ 2) $\frac{ma_1^2}{8t_1^2}(t_2^4 - t_1^4)$ 3) $\frac{ma_1^2}{4t_1^2}(t_2^4 - t_1^4)$ 4) $\frac{ma_1^2}{2t_1^2}(t_2^2 - t_1^2)$

3. Unstretched length of a spring is $\frac{\pi R}{2}$, where R is the radius of smooth vertical circular track in which spring is compressed. One end of the spring is fixed at lowest point A and compressed spring forms an angle of 45° at the centre. A ball of mass m placed at the free end of the spring just rises upto the highest point of the track, find the force constant.



- 1) $\frac{16mg(\sqrt{2}-1)}{\pi^2 R}$ 2) $\frac{16\sqrt{2}mg(\sqrt{2}+1)}{\pi^2 R}$
 3) $\frac{16mg}{\pi^2 R}$ 4) $\frac{16\sqrt{2}mg}{\pi^2 R}$
4. A particle of mass $4m$ which is at rest explodes into 3 fragments. Two of the fragments each of mass m are found to move with a speed V each in mutually perpendicular directions. The total energy released in the process of explosion is:
- 1) $\frac{3}{2}mV^2$ 2) $2mV^2$ 3) $\frac{5}{2}mV^2$ 4) $3mV^2$
5. A square plate, of side a and mass M is placed in $(x-z)$ plane as shown in figure and hinged at A to rotate about x-axis. The plate is allowed to fall upto $(x-y)$ plane. Find the final angular velocity.



- 1) $\sqrt{\frac{24g}{7\sqrt{2}a}}$ 2) $\left[\frac{24g}{7a}\right]^{1/2}$ 3) $\left[\frac{12g}{7a}\right]^{1/2}$ 4) $\left[\frac{12g}{7\sqrt{2}a}\right]^{1/2}$
6. A particle is projected vertically upwards from the surface of the earth (radius R) with a kinetic energy equal to half of the minimum value needed for it to escape. The height to which it rises above the surface of the earth is
- 1) $h = 2.5R$ 2) $h = 2.0R$ 3) $h = 1.5R$ 4) $h = R$

7. A large container of negligible mass and uniform cross-sectional area A has a small hole (of area $a \ll A$) near its side wall at the bottom. The container is open at the top and kept on a smooth horizontal floor. It contains a liquid of density ρ and mass m_0 when liquid starts flowing horizontally at time $t = 0$. Find the speed of container when 75% of the liquid has drained out (Assume the liquid surface remains horizontal throughout the motion)
- 1) $\left[\frac{m_0 g}{A \rho} \right]^{1/2}$ 2) $\left[\frac{4m_0 g}{A \rho} \right]^{1/2}$ 3) $\left[\frac{m_0 g}{2A \rho} \right]^{1/2}$ 4) $\left[\frac{2m_0 g}{A \rho} \right]^{1/2}$
8. A string fixed at both ends is vibrating with 3 antinodes as the possible mode of vibration for which a point(p) at one sixth of its length from one end is a point of maximum displacement. The frequency of vibration in this mode is 100 Hz. What will be the frequency emitted when it vibrates in the next mode such that this point(p) is again a point of maximum displacement?
- 1) 200 Hz 2) 300 Hz 3) 400 Hz 4) 600 Hz
9. A cylinder A of a material of thermal conductivity K_1 is surrounded by a cylindrical shell of inner radius equal to the cylinder. The shell is made of a material of thermal conductivity K_2 . The two ends of the combined system are maintained at two different temperatures. If the effective thermal conductivity of the system is $\frac{4K_1 + 5K_2}{9}$, find the ratio of outer radius of the shell to the radius of the cylinder.
- 1) 2.0 2) 5/2 3) 5/3 4) 3/2
10. In an adiabatic process, $R = \frac{2}{3} C_v$. The pressure of the gas will be proportional to:
- 1) $T^{5/3}$ 2) $T^{5/2}$ 3) $T^{5/4}$ 4) $T^{5/6}$
11. The molar heat capacity for an ideal monoatomic gas is $3R$. If dQ is the heat supplied to the gas and dU is the change in its internal energy, then for the process,
- 1) $dQ = dU$ 2) $dQ = 2dU$ 3) $dQ = -3dU$ 4) $dQ = -\frac{3}{2}dU$
12. In YDSE, bichromatic light of wavelengths 400 nm and 560 nm are used. The distance between slits is 0.1 mm and distance between the plane of the slits and the screen is 1.0 m. The minimum distance between two successive regions of complete darkness is
- 1) 4 mm 2) 5.6 mm 3) 14 mm 4) 28 mm
13. A convex lens is in contact with concave lens. Magnitude of the ratio of their focal lengths is $2/3$. Their equivalent focal length is 30 cm. What are their individual focal lengths (in cm)?
- 1) -75, 50 2) -10, 15 3) 75, 50 4) 10, -15

14. The charges appearing on inner surfaces of a parallel plate capacitor are q_1 and q_2 . The charges appearing on outer surfaces of the capacitor are equal. Find the potential difference between plates of the capacitor, if capacitance is C .

1) $\frac{q_1 + q_2}{2C}$ 2) $\frac{q_1 - q_2}{2C}$ 3) $\frac{q_1 + q_2}{C}$ 4) $\frac{q_1 - q_2}{C}$

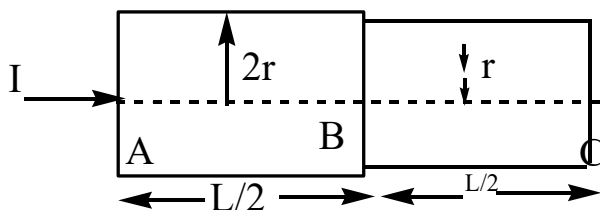
15. Two parallel plate capacitors of capacitances C and $2C$ are connected in parallel and charged to a potential difference V . The battery is then disconnected and the region between the plates of capacitor C is completely filled with a material of dielectric constant K . The potential difference across the capacitors now becomes

1) $\left(\frac{2}{K+1}\right)V$ 2) $\left(\frac{3}{K+1}\right)V$ 3) $\left(\frac{3}{K+2}\right)V$ 4) $\left(\frac{3}{K+3}\right)V$

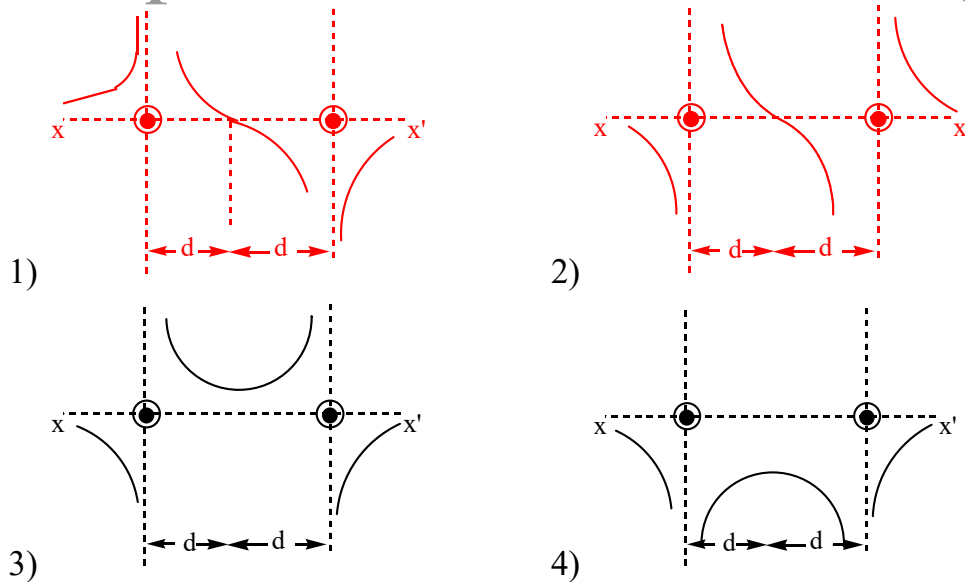
16. A photon collides with a stationary hydrogen atom in ground state inelastically. Energy of the colliding photon is 10.2 eV. After a time interval of the order of micro second another photon collides with same hydrogen atom inelastically with an energy of 15 eV. What will be observed by the detector?

- 1) 2 photon of energy 10.2 eV
 2) 2 photon of energy of 1.4 eV
 3) one photon of energy 10.2 eV and an electron of energy 1.4 eV
 4) one photon of energy 10.2 eV and another photon of 1.4 eV

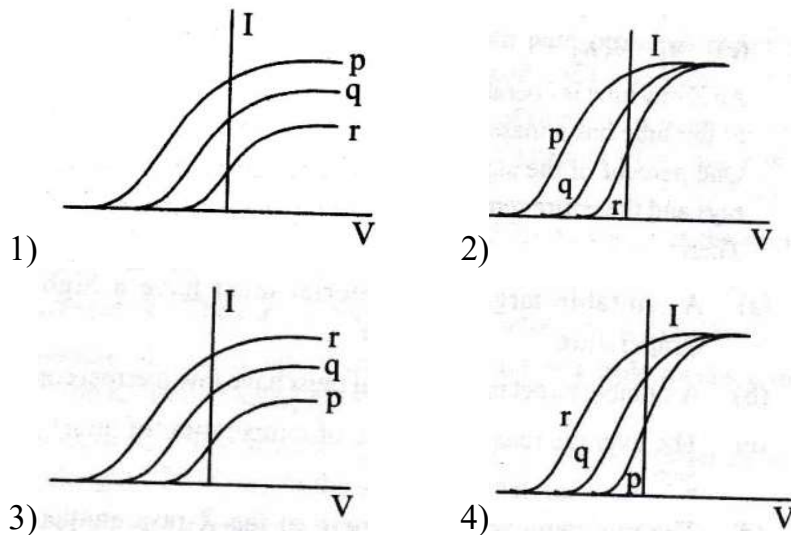
17. Two cylindrical bars of same material with radii r and $2r$ are kept in contact as shown. An electric current I is passed through the bars. Which one of the following is correct?



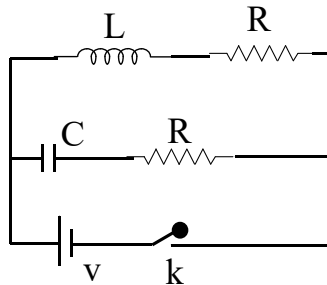
- 1) Heat produced in bar BC is 4 times the heat produced in bar AB
 2) Electric field in both halves is equal
 3) Current density across AB is double that across BC
 4) Potential difference across AB is 4 times that across BC
18. Two long parallel wires are at a distance $2d$ apart. They carry steady equal currents flowing out of the plane of the paper as shown. The variation of magnetic field B along the line XX' is given by:



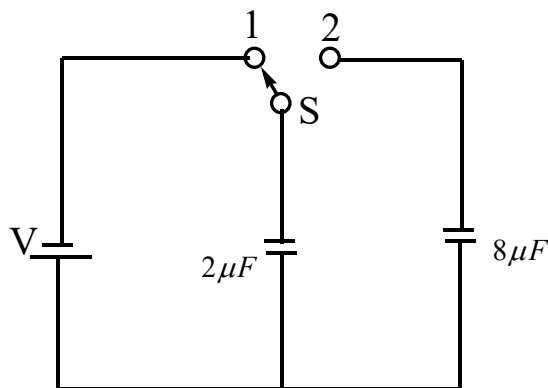
19. Photoelectric effect experiments are performed using three different metal plates p, q and r having work functions $\phi_p = 2.0\text{eV}$, $\phi_q = 2.5\text{eV}$ and $\phi_r = 3.0\text{eV}$ respectively. A light beam containing wavelengths of 550 nm, 450 nm and 350 nm with equal intensities illuminates each of the plates. The correct I-V graph for the experiments is {take $hc = 1240\text{eV nm}$ }



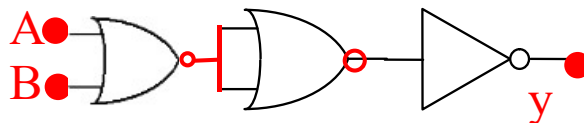
20. The switch is closed at $t = 0$. The current through inductor and capacitor are equal after time $CR \ln(2)$. Then resistance R is equal to



- 1) $\sqrt{L/2C}$ 2) $\sqrt{L/C}$ 3) $\sqrt{\frac{2L}{C}}$ 4) $\left(\frac{L}{C}\right)^{3/2}$
21. A nucleus with mass number 220 initially at rest emits an α -particle. If the Q value of the reaction is 5.5 MeV, calculate the kinetic energy of the α -particle
 1) 4.4 MeV 2) 5.4 MeV 3) 5.6 MeV 4) 6.5 MeV
22. When 100 V dc is applied across a coil, a current of 1 A flows through it. When 100 V ac of 50 Hz is applied across the same coil, only 0.5 A flows. The resistance and inductance of the coil are (take $\pi^2 = 10$)
 1) $50\Omega, 0.3H$ 2) $50\Omega, \sqrt{(0.3)}H$
 3) $100\Omega, 0.3H$ 4) $100\Omega, \sqrt{(0.3)}H$
23. The near point of a person is 50 cm and the far point is 1.5 m. The spectacles required for reading purpose and for seeing distant objects are respectively
 1) $+2D, -\frac{2}{3}D$ 2) $+\frac{2}{3}D, -2D$ 3) $-2D, +\frac{2}{3}D$ 4) $-\frac{2}{3}D, +2D$
24. A capacitor of $2\mu F$ is charged as shown in the diagram. When the switch S is turned to position 2, the percentage of its energy dissipated is:

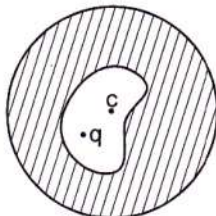


- 1) 80% 2) 0% 3) 20% 4) 75%
25. The given network is equivalent to:



- 1) OR gate 2) NOR gate 3) NOT gate 4) AND gate
26. If θ_1 and θ_2 be the apparent angles of dip observed in two vertical planes at right angles to each other on either side of magnetic meridian, then the true angle of dip θ is given by
 1) $\tan^2 \theta = \tan^2 \theta_1 + \tan^2 \theta_2$ 2) $\cot^2 \theta = \cot^2 \theta_1 - \cot^2 \theta_2$
 3) $\tan^2 \theta = \tan^2 \theta_1 - \tan^2 \theta_2$ 4) $\cot^2 \theta = \cot^2 \theta_1 + \cot^2 \theta_2$

27. A message signal of frequency 10 kHz and peak voltage of 10 volts is used to modulate a carrier of frequency 1 MHz and peak voltage of 20 volts. The side bands are
 1) 1010 kHz and 990 kHz 2) 1000 kHz and 900 kHz
 3) 10 kHz and 1 MHz 4) 505 kHz and 445 kHz
28. The figure shows a charge q placed inside a cavity in an uncharged conductor. Now if an external electric field is switched on



- 1) Only induced charge on outer surface will redistribute
 2) Only induced charge on inner surface will redistribute
 3) Both induced charge on outer and inner surface will redistribute
 4) Force on charge q placed inside the cavity will change.
29. Match the entries of column I with corresponding entries of column 2. {where m is magnification produced by the mirror for real object}

Column 1		Column 2	
A)	$m = -2$	a)	Convex mirror
B)	$m = -\frac{1}{2}$	b)	Concave mirror
C)	$m = +2$	c)	Real image
D)	$m = +\frac{1}{2}$	d)	Virtual image

- 1) $A \rightarrow c$ and d ; $B \rightarrow b$ and d; $C \rightarrow b$ and c; $D \rightarrow a$ and d
 2) $A \rightarrow b$ and c; $B \rightarrow b$ and c; $C \rightarrow b$ and d; $D \rightarrow a$ and d
 3) $A \rightarrow a$, and c; $B \rightarrow a$ and d; $C \rightarrow a$ and b; $D \rightarrow c$ and d
 4) $A \rightarrow a$ and d; $B \rightarrow b$ and d; $C \rightarrow b$ and d; $D \rightarrow a$ and d
30. A long solenoid of diameter 0.1 m has 2×10^4 turns per meter. At the centre of the solenoid, a coil of 100 turns and radius 0.01 m is placed with its axis coinciding with the solenoid axis. The current in the solenoid reduces at a constant rate to 0A from 4A in 0.05 s. If the resistance of the coil is $10\pi^2 \Omega$. The total charge flown through the coil during this time is:
 1) $16\mu C$ 2) $32\mu C$ 3) $16\pi\mu C$ 4) $32\pi\mu C$

MATHS

31. In $[-2\pi, 2\pi]$ the number of solution of $\ln(\sin x) = x$ is
 1) 0 2) 2 3) 4 4) 8

32. If $\frac{\cos x}{\cos \theta} + \frac{\sin x}{\sin \theta} = \frac{\cos y}{\cos \theta} + \frac{\sin y}{\sin \theta} = 1$, where $x - y \neq n\pi, n \in \mathbb{Z}$, then

$$\frac{\cos x \cos y}{\cos^2 \theta} + \frac{\sin x \sin y}{\sin^2 \theta} =$$
1) $1/2$ 2) 1 3) -1 4) $3/2$
33. If $x = \tan^{-1}(\tan(-6))$, $y = \cot^{-1}(\cot 4)$, then $2x + 3y =$
1) 6 2) $6 - \pi$ 3) $\pi - 6$ 4) π
34. With usual notation in triangle ABC, b, c, B are given such that $b < c$. If a_1, a_2 are two positive values of third side satisfying $(a_1 - a_2)^2 + (a_1 + a_2)^2 \tan^2 B = kb^2$, then $k =$
1) 4 2) 2 3) 1 4) 3
35. The angle of elevation of a tower CD at a place A due south of it is 60° and at a place B due west of A, the elevation is 30° . If $AB = 3\text{KM}$, height of the tower is
1) $\frac{2}{3\sqrt{6}}\text{KM}$ 2) $\frac{2\sqrt{2}}{3\sqrt{3}}\text{KM}$ 3) $\frac{3\sqrt{3}}{2\sqrt{2}}\text{KM}$ 4) $\frac{3\sqrt{6}}{2}\text{KM}$
36. If the roots of $x^2 + a(x-1) - b(x+1) + 1 = 0$ are real and unequal for all real values of b then a belongs to
1) $(1, \infty)$ 2) $(-\infty, 1)$ 3) $(-1, \infty)$ 4) $(-\infty, -1)$
37. If $x = \sqrt{1 + \frac{1}{1^2} + \frac{1}{2^2}} + \sqrt{1 + \frac{1}{2^2} + \frac{1}{3^2}} + \dots \dots \sqrt{1 + \frac{1}{2019^2} + \frac{1}{2020^2}}$ then
 $4040(2020 - x) =$
1) 1 2) 2 3) -1 4) -2
38. If $\left(4^p\right)^{\frac{1}{q}} - 3^{\frac{2p-q}{2q}} = \left(3^{2p+q}\right)^{\frac{1}{2q}} - \left(2^{2p-q}\right)^{\frac{1}{q}}$ -such that p, q are positive in tangents and greatest common divisor of p, q is 1, then $3p - 2q =$
1) 0 2) -7 3) 7 4) 5
39. The number of natural numbers which are smaller than 2×10^8 and are divisible by 3 using the digits 0, 1, 2 only is a four digit number $abcd$. Then $(a + c) - (b + d) =$
1) 5 2) 6 3) 4 4) 3
40. The digit in the 10's place of $7^{14^{14}}$ is
1) 4 2) 0 3) 2 4) 5

41. Let $P(E)$ denotes probability of an event E, If $P(B) = \frac{3}{4}$,

$$P(A \cap B \cap \bar{C}) = \frac{1}{3}, P(\bar{A} \cap B \cap \bar{C}) = \frac{1}{3}, \text{ then } P(B \cap C) =$$

- 1) $\frac{1}{4}$ 2) $\frac{5}{12}$ 3) $\frac{1}{12}$ 4) $\frac{1}{6}$

42. If the length of the latus rectum of the parabola

$$289(x-3)^2 + 289(y-1)^2 = (15x-8y+13)^2 \text{ is } k, \text{ then } [k] =$$

- 1) 5 2) 2 3) 3 4) 11

43. Two ellipses $\frac{x^2}{\cos^2 \alpha} + \frac{y^2}{\sin^2 \alpha} = 1, \frac{x^2}{\sin^2 \alpha} + \frac{y^2}{\cos^2 \alpha} = 1, \alpha \in \left(0, \frac{\pi}{4}\right)$ intersect at 4

points P, Q, R, S. If two statements given as

Statement-I: PQRS is a square of side $\sin 2\alpha$

Statement-II: Eccentricities of two ellipse are equal, then

- 1) Both are true and II is correct explanation of I
2) Both are true and II is not correct explanation of I
3) I is true, II is false
4) I is false, II is true

44. Let $z_1 = 6 + i, z_2 = 4 - 3i$ and z be a complex number such that $\arg\left(\frac{z - z_1}{z_2 - z}\right) = \frac{\pi}{2}$

.If two statements are given by

$$\text{Statement-I: } (z - (6 + i))(\bar{z} - (4 + 3i)) + (z - (4 - 3i))(\bar{z} - (6 - i)) = 0$$

Statement-II: Z satisfies $|z - (5 - i)| = 5$, Then

- 1) Both are true and II is correct explanation of I
2) Both are true and II is not correct explanation of I
3) I is true, II is false
4) I is false, II is true

45. A circle touches the line $2x + 3y + 1 = 0$ at the point $(1, -1)$ and is orthogonal to the circle which has the line segment having end points $(0, -1)(-2, 3)$ as diameter is given by $x^2 + y^2 + 2gx + 2fy + c = 0$. Then, which of the following is not correct?

- 1) $g = -\frac{5}{2}$ 2) $2g + 2f + c = -7$

3) $f = -\frac{5}{4}$

4) $c = -\frac{1}{2}$

46. Let $P: 2$ is a prime number as $q: \sqrt{3}$ is rational be two propositions. Then the truth values of biconditional Statements

a) $p \Leftrightarrow q$

b) $\sim p \Leftrightarrow q$

c) $\sim q \Leftrightarrow p$

d) $\sim q(>) \sim q$ in order are given by

1) FTFT

2) FTTF

3) FFTT

4) TFTF

47. If $A = (2, 5), B = (5, 11)$ and a point P moves such that internal bisector of $\angle APB$ passes through $(4, 9)$. The maximum area of $\triangle APB$ is

1) $\frac{45}{4} \text{ sq. units}$

2) $\frac{15}{2} \text{ sq. units}$

3) 15 sq. units

4) $\frac{45}{2} \text{ sq. units}$

48. Let 'L' be the projection of the line $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z-3}{4}$ on the plane

$$x + 2y + z = 9$$

Then, which of the following is not correct?

1) $\left(\frac{1}{2}, \frac{17}{4}, 0\right)$ is a point on L

2) $\left(\frac{3}{2}, \frac{15}{4}, 0\right)$ is a point on L

3) $\left(\frac{13}{6}, \frac{4}{3}, \frac{25}{6}\right)$ is a point on L

4) Direction ratios on L are $(4, -7, 10)$

49. The volume of triangular prism whose adjacent sides are $\vec{a}, \vec{b}, \vec{c}$ each of magnitude 4

units and each is inclined at an angle $\frac{\pi}{3}$ with other two is

1) $16\sqrt{2}$

2) $16\sqrt{3}$

3) $8\sqrt{3}$

4) $8\sqrt{2}$

50. If $a \neq p, b \neq q, c \neq r$ and $\begin{vmatrix} p & b & c \\ a & q & c \\ a & b & r \end{vmatrix} = 0$, then $\frac{p}{a-p} + \frac{q}{b-q} + \frac{r}{c-r} = \underline{\hspace{2cm}}$

1) 1

2) -1

3) 2

4) -2

51. If all the linear functions defined from $[-1, 1]$ onto $[0, 3]$ intersect at (p, q) , then $2019p + 2020q =$
 1) 3030 2) 4038 3) -1 4) 4040
52. If $2f(\sin x) + \sqrt{2}f(\cos x) = \tan x, (x > 0)$, then $\lim_{x \rightarrow 1} \sqrt{1-x} f(x) =$
 1) $\sqrt{2}$ 2) $\frac{1}{\sqrt{2}}$ 3) $-\sqrt{2}$ 4) $-\frac{1}{\sqrt{2}}$
53. The tangent at any point on the curve $x^{2/3} + y^{2/3} = a^{2/3}$ meets the axes at P and Q. The locus of midpoint of PQ is ____
 1) $x^2 + y^2 = \frac{a^2}{2}$ 2) $x^2 + y^2 = \frac{a^2}{3}$ 3) $x^2 + y^2 = \frac{a^2}{4}$ 4) $x^2 + y^2 = a^2$
54. Let $p(x)$ be a polynomial of least degree having maximum value 6 at $x=1$ and minimum value 2 at $x=3$. then $p(2) + p'(0) =$
 1) 9 2) 11 3) 12 4) 13
55. The value of k for which the equation $x^3 - 3x + k = 0$ has two distinct roots is (0,1) is
 1) in the interval $(-2, -1) \cup (1, 2)$ 2) in the interval $(-1, 1)$
 3) in the interval $(0, 1)$ 4) not existing
56. $f(x)$ is a quadratic function such that $f(0) = 1, f(-1) = 4$. If $\int \frac{f(x) dx}{x^2(x+1)^2}$ is a rational function, then $f(10) =$ ____
 1) 512 2) 521 3) 432 4) 324
57. The area of the region bounded by the curve $y = e^x$ and the lines $x = 0, y = e$ is
 1) 1 2) $e-1$ 3) $e - \frac{1}{e}$ 4) $e^e - 1$
58. If $\sin y \frac{dy}{dx} = \cos y (1 - x \cos y)$ and $y(0) = \frac{\pi}{3}$, then $y(\log 2) =$
 1) $\tan^{-1}(2 + \log 2)$ 2) $\sec^{-1}(2 + \log 2)$
 3) $\sec^{-1}(3 + \log 2)$ 4) $\tan^{-1}(3 + \log 2)$

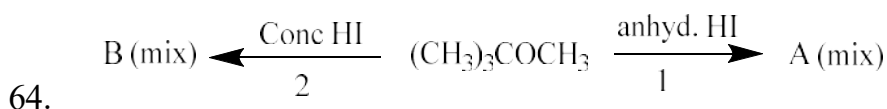
59. If σ is the standard deviation of the numbers 10, 15, 25, 30, x, then the value of x for which σ is least is
- 1) 5 2) 20 3) 30 4) 80
60. For real numbers x and y, $xRy \Leftrightarrow x - y + \sqrt{2}$ is an irrational number. Then the relation R is
- 1) symmetric and transitive 2) symmetric but not transitive
3) transitive but not symmetric 4) neither symmetric nor transitive

CHEMISTRY

61. By how much would the oxidising power of the $\text{MnO}_4^- / \text{Mn}^{2+}$ couple change if the H^+ ions concentration is decreased 100 times?
- 1) Increases by 189 mV 2) Decreases by 189 mV
3) Increases by 19 mV 4) Decreases by 19 mV
62. In an fcc arrangement of A and B atoms, where A atoms are at the corners of the unit cell, B atoms at the face centers, two atoms are missing from two corners in each unit cell, then the simplest formula of the compound is
- 1) A_7B_6 2) A_6B_7 3) A_7B_{24} 4) AB_4
63. The enthalpy of combustion of $\text{H}_2(\text{g})$ at 298 K to give H_2O is -298 kJ mol^{-1} and bond enthalpies of H-H and O = O are 433 kJ mol^{-1} and 492 kJ mol^{-1} , respectively. The mean bond enthalpy of O-H is :

(ΔH_{vap} of $\text{H}_2\text{O} = 40.65 \text{ kJ / mol}$)

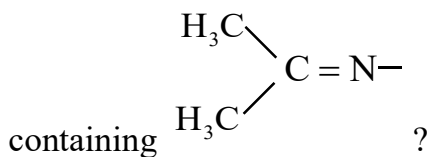
- 1) 464 kJ mol^{-1} 2) $488.5 \text{ kJ mol}^{-1}$ 3) 232 kJ mol^{-1} 4) -232 kJ mol^{-1}



- 1) A and B are identical mixtures of CH_3I and $(\text{CH}_3)_3\text{C}-\text{OH}$
2) A and B are identical mixtures of CH_3OH and $(\text{CH}_3)_3\text{C}-\text{I}$
3) A is mixture of CH_3I and $(\text{CH}_3)_3\text{C}-\text{OH}$; B is a mixture of CH_3OH and $(\text{CH}_3)_3\text{C}-\text{I}$

4) None of these

65. Which of the following will react at faster rate with acetone to give a product



1) $\text{C}_6\text{H}_5\text{NH}_2$

2) $(\text{CH}_3)_3\text{N}$

3) $\text{C}_6\text{H}_5 \cdot \text{NHC}_6\text{H}_5$

4) $\text{C}_6\text{H}_5\text{NHNH}_2$

66. A negatively charged sol can be formed by peptizing a solution of

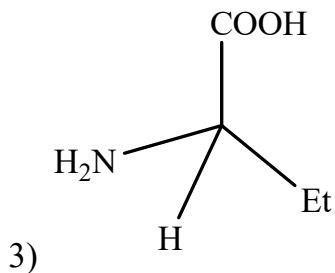
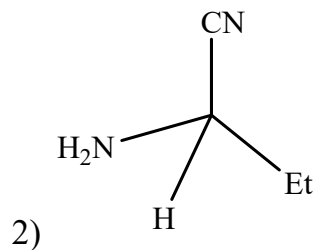
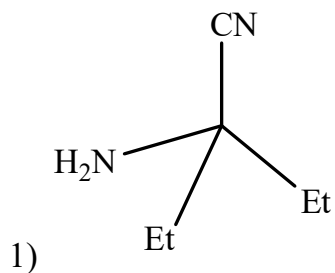
1) AgI with AgNO_3

2) AgI with KI

3) $\text{Fe}(\text{OH})_3$ with FeCl_3

4) Any of these

67. What is A in the following reaction?



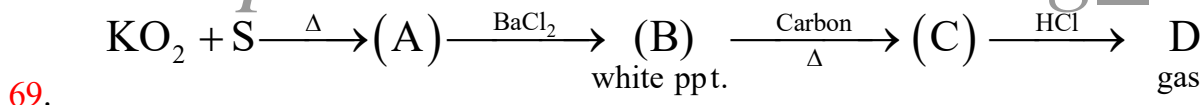
4) No reaction takes place

68. Among the following species identify the isostructural pairs: NF_3 , NO_3^- , BF_3 ,

H_3O^+ , HN_3

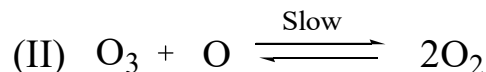
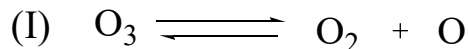
1) $[\text{NF}_3, \text{NO}_3^-]$ and $[\text{BF}_3, \text{H}_3\text{O}^+]$ 2) $[\text{NF}_3, \text{HN}_3]$ and $[\text{NO}_3^-, \text{BF}_3]$

3) $[\text{NF}_3, \text{H}_3\text{O}^+]$ and $[\text{NO}_3^-, \text{BF}_3]$ 4) $[\text{NF}_3, \text{H}_3\text{O}^+]$ and $[\text{HN}_3, \text{BF}_3]$



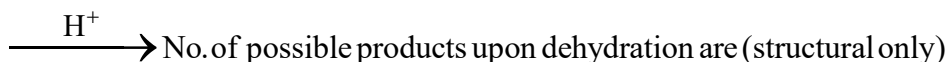
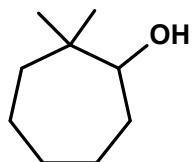
- 1) (B) is BaSO_4 and (D) is H_2S 2) (A) is BaS and (D) is SO_2
 3) (C) is BaS and (D) is SO_2 4) (A) is K_2SO_3 and (B) is BaS

70. The reaction; $2\text{O}_3 \rightarrow 3\text{O}_2$, is assigned the following mechanism.



The rate law of the reaction will therefore be

- 1) $r \propto [\text{O}_3]^2 [\text{O}_2]$ 2) $r \propto [\text{O}_3]^2 [\text{O}_2]^{-1}$
 3) $r \propto [\text{O}_3]$ 4) $r \propto [\text{O}_3] [\text{O}_2]^{-2}$



71.

- 1) 3 2) 6 3) 7 4) 4

72. How many chlorine atoms can you ionize in the process $\text{Cl} \rightarrow \text{Cl}^+ + e^-$, by the energy liberated from the following process? $\text{Cl} + e^- \rightarrow \text{Cl}^-$ for 6×10^{23} atoms. Given electron affinity of $\text{Cl} = 3.61 \text{ eV/atom}$, and IP of $\text{Cl} = 17.422 \text{ eV/atom}$.

- 1) 1.24×10^{23} atoms 2) 9.82×10^{20} atoms
 3) 2.02×10^{15} atoms 4) None of these

73. 0.5 g of fuming H_2SO_4 (oleum) is diluted with water. This solution is completely neutralised by 26.7 ml of 0.4 N NaOH . The percentage of free SO_3 in the sample is

- 1) 30.6% 2) 40.6% 3) 20.6% 4) 50%

74. Arrange the elements with the following electronic configurations in the increasing order of electron affinity

(i) $1S^2 2S^2 2P^3$ (ii) $1S^2 2S^2 2P^4$ (iii) $1S^2 2S^2 2P^3$ (iv) $1S^2 2S^2 2P^6 3S^2 3P^4$

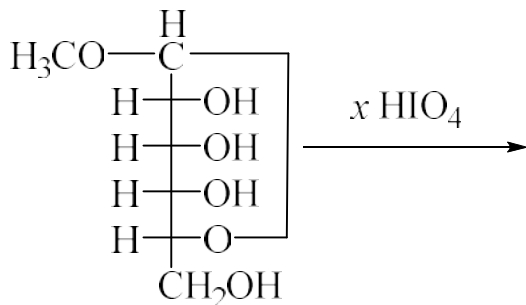
1) $i < ii < iii < iv$

2) $iv < iii < ii < i$

3) $iii < ii < iv < i$

4) $ii < iii < i < iv$

75. What is the maximum value of x ?



1) 1

2) 2

3) 3

4) 4

76. Sec. butyl chloride will undergo alkaline hydrolysis in the polar solvent by

1) S_N^2

2) S_N^1

3) S_N^1 and S_N^2

4) None of these

77. A solution of sodium sulphate in water is electrolysed using inert electrodes. The products at cathode and anode are, respectively,

1) H_2, O_2

2) O_2, H_2

3) O_2, Na

4) O_2, SO_2

78. Which of the following compounds yields only one product on monobromination?

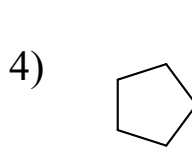
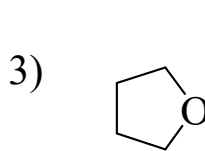
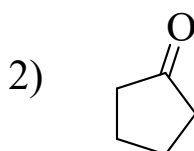
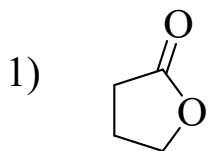
1) Neopentane

2) Toluene

3) Phenol

4) Aniline

79. $HOCH_2CH_2CH_2COOH$ on reaction with H^+ forms



80. 2 moles of iso-pentylene on reaction with A gives compound B, which on treatment with 1-phenyl-2-butyne followed by H_2O_2 oxidation results in C. A is in-turn generated by treating BCl_3 with $LiAlH_4$. Compound C gives

1) Tollen's reagent

2) Iodoform test

3) Fehling's test

4) 1 & 3

81. How many enantiomers pairs are produced by the monochlorination of 2-methylbutane?
 1) 1 2) 2 3) 4 4) 6
82. The correct order among the following is
 1) $\text{ClO}_4^- > \text{BrO}_4^- > \text{IO}_4^-$ (basicity)
 2) $\text{ClO}^- > \text{ClO}_2^- > \text{ClO}_3^- > \text{ClO}_4^-$ (delocalization of negative charge)
 3) $\text{ClO}^- > \text{ClO}_2^- > \text{ClO}_3^- > \text{ClO}_4^-$ (charge density)
 4) $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3$ (delocalization of lone pair electron)
83. The number of isomers possible for C_4H_8 are
 1) 4 2) 3 3) 5 4) 6
84. High concentration of fluoride is poisonous and harmful to bones and teeth at levels over
 1) 1 ppm 2) 3 ppm 3) 5 ppm 4) 10 ppm
85. Mole fraction of a non electrolyte in aqueous solution is 0.07. If K_f is $1.86^\circ \text{ mol}^{-1} \text{ kg}$, depression in f.p, ΔT_f is,
 1) 0.26° 2) 1.86° 3) 0.13° 4) 7.78°
86. The pH of a buffer solution changes from 6.20 to 6.17 when 0.003 mole of acid is added to 500 mL of the buffer. The buffer capacity of the system is, therefore,
 1) 0.1 2) 0.3 3) 0.2 4) 0.4
87. The volume occupied by 2 mol of N_2 at 200K and 10.1325 MPa pressure is (Given that $\frac{P_C V_C}{RT_C} = \frac{3}{8}$ and $\frac{P_r V_r}{T_r} = 2.21$), where P_r, V_r & T_r are reduced pressure, reduced volume & reduced temperature.
 1) 20.7 cm^3 2) 544 cm^3 3) 272.0 cm^3 4) 136 cm^3
88. Choose the correct code to identify (X), (Y) and 'Z' in each case for the changes indicated.
- (i) $\text{CrO}_2\text{Cl}_2 \xrightarrow{\text{KOH}} (\text{X}) \xrightarrow{\text{Conc. H}_2\text{SO}_4} (\text{Y}) \xrightarrow{\text{AgNO}_3} (\text{Z})$
- (ii) $\text{CrCl}_3 (\text{aq}) \xrightarrow[\text{NaOH}]{\text{excess}} (\text{X}) \xrightarrow[\text{H}_2\text{O, boil}]{\text{Na}_2\text{O}_2} (\text{Y}) \xrightarrow[\text{acetate}]{\text{lead}} (\text{Z})$
- (iii) $\text{ZnSO}_4 (\text{aq}) \xrightarrow{\text{Na}_2\text{CO}_3} (\text{X}) \xrightarrow{\Delta} (\text{Y}) \xrightarrow[\text{nitrate, } \Delta]{\text{cobalt}} (\text{Z})$
- (iv) $\text{CuCl}_2 (\text{aq}) \xrightarrow[\text{H}_2\text{S}]{\text{NH}_4\text{OH}} (\text{X}) \xrightarrow[\Delta]{\text{HNO}_3} (\text{Y}) \xrightarrow[\text{excess}]{\text{KCN}} (\text{Z})$

- 1) $X = K_2CrO_4; Y = K_2Cr_2O_7; Z = Ag_2CrO_4$
- 2) $X = Cr(OH)_3; Y = Na_2CrO_4; Z = PbCrO_4$
- 3) $X = ZnCO_3; Y = ZnO; Z = CoZnO_2$
- 4) $X = CuS; Y = Cu(NO_3)_2; Z = K_3[Cu(CN)_4]$

89. Which of the following statements is incorrect?

- 1) The order of splitting energy is $PtCl_4^{2-} > PdCl_4^{2-} > NiCl_4^{2-}$
- 2) $[Co(NH_3)_6]^{3+}$ is colourless whereas $[Ni(H_2O)_6]^{2+}$ is coloured.
- 3) $[M(en)(gly)]^{n+}$ will exhibit geometrical isomerism.
- 4) The magnetic moment of $K_3[Fe(CN)_6]$ is $\sqrt{3}$ B.M.

90. Which of the following reactions does not occur in Bessemer convertor in the extraction of copper from chalcopyrites?

- 1) $2CuFeS_2 + O_2 \longrightarrow Cu_2S + 2FeS + SO_2$
- 2) $FeO + SiO_2 \longrightarrow FeSiO_3$
- 3) $2FeS + 3O_2 \longrightarrow 2FeO + 2SO_2$
- 4) $Cu_2S + 2Cu_2O \longrightarrow 6Cu + SO_2$

KEY SHEET

1	1	2	2	3	2	4	1	5	1
6	4	7	4	8	2	9	4	10	2
11	2	12	4	13	4	14	2	15	3
16	3	17	1	18	2	19	1	20	2
21	2	22	4	23	1	24	1	25	2
26	4	27	1	28	1	29	2	30	2
31	2	32	3	33	4	34	1	35	3
36	1	37	2	38	4	39	1	40	2
41	3	42	1	43	2	44	3	45	4
46	2	47	3	48	2	49	1	50	4
51	1	52	2	53	3	54	4	55	4
56	2	57	1	58	3	59	2	60	4
61	2	62	4	63	2	64	3	65	4
66	2	67	1	68	3	69	1	70	2
71	2	72	1	73	3	74	3	75	2
76	2	77	1	78	1	79	1	80	2
81	2	82	3	83	4	84	4	85	4
86	3	87	3	88	4	89	3	90	1